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Shannon Entropy in Artificial Intelligence and Its Applications Based on Information Theory

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***Abstract*-- In the 21st century, the era of communication and technology, entropy arises as an application of information theory in safe data transfer, lossless data compression, data mining, machine learning, and classification. *Shannon's entropy formula* is a basic information theory measuring tool. This work applies Shannon's entropy formula to information theory. First, information and entropy were computed in a fair and fraudulent coin flip game and compared. The fair coin's entropy was more significant than the counterfeit coin's, indicating that entropy measures system uncertainty. With entropy in information theory, effective communication and data storage utilizing few bits have improved. The second application computed the lowest number of bits necessary to code a data group's characters using the entropy formula. Considering several elements, such as the change in information capacity through time and the development of systems to manage and steer it, the measurement, quality, and significance of information are better understood. Claude E. Shannon's Theory of Knowledge also advanced controlled knowledge management. Shannon's ideas inspired the notion of entropy, which later approaches utilized to measure information uncertainty. Entropy, which measures information disorder, is significant in many disciplines. In the last application, the entropy formula was utilized to find the root node in creating the Decision Tree from affected learning regions. Shannon entropy has applications in information theory. In recent years, Artificial Intelligence (AI) has progressed dramatically with Big Data and Deep Learning, and AI can now handle more extensive datasets. This research examined the AI notion of entropy. While the idea and concept are the same, the methodological implementation differs. The study aims to help build novel entropy and AI approaches.**

***Keywords*- Entropy, Information Theory, Information Gain, Artificial Intelligence, Decision Tree.**

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I. Introduction

The notion of entropy, which has numerous applications in physics, statistics, engineering, and social sciences, is derived from thermodynamics. Entropy, which is dispersed throughout the cosmos, is the thermal energy of a system that cannot be turned into mechanical work in thermodynamics [1]. For instance, not all of a ball's potential energy can be turned into kinetic energy when it falls down a hill; instead, part of it is released into the cosmos as heat energy. *Entropy* is the energy released as heat and dispersed erratically across the cosmos [2]. According to the second rule of thermodynamics, the system's entropy must either increase or decrease throughout any operation. Because entropy tends to rise, all processes in the cosmos experience a transition from order to chaos. Entropy in physical systems is often referred to as the system's degree of uncertainty [3].

The communication process exhibits the entropic direction of the cosmos, which shifts from order to chaos. Claude E. Shannon identified the distortion experienced during message transmission via a channel as entropy [4]. Entropy is a measurement of the amount of information included in a message, according to Shannon, who is regarded as the father of information theory. The earliest research on this topic focused on creating an efficient communication process. This research on information transfer formed the foundation for Shannon's entropy formula [5].

Harry Nyquist conducted the first investigation on information transfer in 1924 [6]. His paper "Certain Factors Affecting Telegraph Speed" described how to deliver telegraph messages as quickly and accurately as possible. Can information be measured? Was the main subject of Ralph Hartley's 1928 paper titled "Transmission of Information" [7]. After studying this topic, Hartley created the formula shown below:

$$\mathrm{H} = \log_s n \tag{1}$$

Where H stands for information, n for the number of symbols, and s for the number of symbols used for each choice.

We may infer from this equation that the message space's logarithm provides us with the amount of information.

Let us use an illustration to try to make understand this Hartley formula. Let us try to predict a number that is between 0 and 50. Let us Fig. out how many "yes" and "no" questions we must ask to be guessed. We could inquire, "Is this number 1?", "Is this number 2?" or "Is this number 3?" but this would be illogical. Considering that, at most, we are aware of the solution to the first question's digit, and, at worst, we can locate it in question 50. Instead, if we phrase the question as, "Is this number less than 25?" we may rule out half of the potential answers, and then by halving the remaining questions in the subsequent inquiry, we can arrive at the solution with fewer inquiries. According to Hartley's technique, the average number of questions is determined as follows: log2 50 = several questions. If 50 questions are asked, the base of the logarithm, in this case, is 2, which stands for either "yes" or "no." In this instance, the outcome is log2 50 = 5.64, indicating that 5.64 is the average number of questions. Hartley says we should ask "pieces" of inquiries in this situation. Thus, one query equals one bit, explains Hartley. We may infer that 5.64 bits can be used to store this information.

Hartley's formula laid the path for developing the theory of knowledge. The 1948 publication of "A Mathematical Theory of Communication" by Claude E. Shannon [8] served as the foundation for the development of modern information theory. In this study, Shannon created a mathematical model of communication in which he demonstrated how successful and excellent communication may be, laying the groundwork for information theory in the process [8]. Shannon later refined it to identify the limitations of signal processing activities, including data compression, storage, and secure transmission. Here is Shannon's description of the communication system:

The system that creates a message or series of messages for transmission to the receiving device is identified as the information source. The transmitter is the component that transforms the information source's message into a format suitable for channel transmission. The signal is the term used to describe the format in which a message might be sent across a channel. The signal travels from the transmitter to the receiver over a medium called a channel [9]. The component known as the receiver reverses what the transmitter accomplishes by turning the signal back into a message. The target displays the most recent point the message has reached. Due to the environment's properties, several changes happen while the signal is sent over the channel. Noise is the name for these unwanted shifts. The message coming from the source and the message being received by the target will differ if there is noise. Corruption occurs more frequently in communications that are transferred from sender to recipient. Messages, on the other hand, have a very slim chance of getting to their intended receiver unaltered. Shannon demonstrated that it is feasible to send information via a noisy channel with relatively little error and described this time-increasing deterioration in transmitted messages as entropy. The information theory proposed by Shannon offers a solution to a crucial issue in communication technologies: the quickest means to deliver accurate information [10].

This research aims to demonstrate how helpful entropy is in areas of application like data storage and transmission, varieties of entropy used in Artificial Intelligence approaches its contributions to the growth of information and communication technologies, and its usage in the construction of decision trees in the artificial learning space. The Shannon entropy is defined, and several examples of its applications are provided in the following section. Most of advance research work done in current days should provide excellent direction to the future researcher [11-17].

II. Entropy

Entropy is utilized to quantify disorder in every domain where disorder is specified. It is described as "The number of bits expressing the level of uncertainty in a data source" in information theory [18]. The letter H stands for entropy in information theory. The disorder of information in a medium is assessed using the letter H. At this moment, the disorder is eliminated by the information flowing consistently. In the entropy formula, n stands for the total amount of data and pi for the likelihood that the data are relevant [19].

$$H = -\sum_{i=1}^{n} p_i \log(p_i) \tag{2}$$

According to the formulae and definitions described above, entropy is greatest when the two probabilities are equal and assuming two states with p and $1-p$ probability [20]. Fig. 1 illustrates that the level of uncertainty peaks at $p = 0.5$. Uncertainty starts to vanish when a situation's likelihood rises.

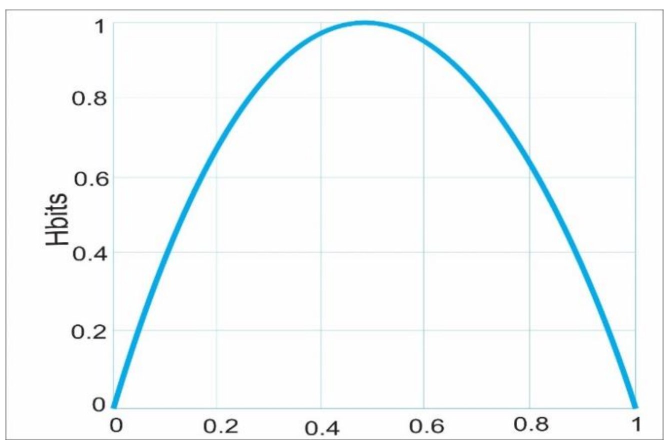

Fig. 1: Example of entropy change in the H-bit data amount of two states with the probability p and 1-p)

A. Properties of Entropy

Entropy's framework is described in terms of a few characteristics when it is defined in information theory. Below is a description of each of these characteristics.

Feature 1: The probability of every state except one is zero if H=0. All possibilities exist in a condition that is still. If not, H is positive. (For instance, brutal dice keep appearing) [21].

Feature 2: When all pi is equal (all 1/n) for a given n, H is at its maximum and equal to log (n). Additionally, this instance is the most illogical [22].

Feature 3: Assume that two occurrences, x, and y, occur and that m and n are the relevant m and n, respectively.

- The probability of co-occurrence i for the first and j for the second is $p(i,j)$. The entropy of this co-occurrence:

$$H(x,y) = -\sum_{i,j} p(i,j)\log p(i,j) \tag{3}$$

$$H(x) = -\sum_{i,j} p(i,j)\log \sum_j p(i,j) \tag{4}$$

$$H(y) = -\sum_{i,j} p(i,j)\log \sum_j p(i,j) \tag{5}$$

- The occurrences must be independent in order for the disparity to be reflected. The uncertainty of a shared occurrence is now considered equal to or lower than the sum of the individual uncertainties [23].

$$H(x,y) \leq H(x) + H(y) \tag{6}$$

$$p(i,j) = p(i)p(j) \tag{7}$$

Feature 4: Any change in the probability of pi in a problem being equal increases H.

- For example, let $p1 < p2$. Increasing $p1$ is a step towards making it converge or even equal to $p2$. Entropy takes its maximum value when the probability of two states reaches ½.

Feature 5: Suppose there are x and y events as in the third property.

- The conditional probability of two events based on the value of i for x and j for y is defined as $pi\ (j)$, and the formula is [24]:

$$p_i(j) = \frac{p(i,j)}{\sum_j p(i,j)} \tag{8}$$

- The conditional entropy of y is $Hx(y)$ as the mean entropy of y for each value of x weighted by the probability of obtaining x. This equation measures how uncertain y is on the mean when we know x.

$$H_x(y) = -\sum_{i,j} p(i,j)\log p_i(j) \tag{9}$$

$$H_x(y) = -\sum_{i,j} p(i,j)\log p(i,j) + \sum_{i,j} p(i,j)\log \sum_j p(i,j) \tag{10}$$

$$H_x(y) = H(x,y) - H(x) \tag{11}$$

$$H(x,y) = H(x) + H_x(y) \tag{12}$$

- When x is known, the uncertainty of the x, y joint event is equal to the sum of the uncertainties of x and y.

Feature 6: The following structure is created using the formulae outlined in the third and fifth features [25].

$$H(x) + H(y) \geq H(x,y) = H(x) + H_x(y) \tag{13}$$

- The following inequality can be defined as a result.

$$H(y) \geq H_x(y) \tag{14}$$

- As a result, it may be claimed that as x becomes more and more known, y's level of uncertainty never rises. Unless x and y are separate occurrences, it will even decline [26].

III. The Use of Entropy in Artificial Intelligence

In each information criterion approach, entropy arises uniquely. Cross-Entropy is the most common type of entropy employed in artificial intelligence methods. This section provides a full explanation of cross-entropy and the artificial intelligence methods that employ it [27].

A. Cross-Entropy

In Information Theory, cross-entropy is defined above by Feature -3. The primary purpose here is to get the average number of bits needed to encode an event from q if we use the wrong encoding scheme q instead of p. Evaluation of the event between two states based on disorder is called cross-entropy [28] and is represented by the formula below.

$$H(p,q) = E_p[-\log(q)]\& = -\int_x p(x)\log(q(x))dx \tag{15}$$

Entropy-based studies are also conducted to compare the predicted and required information in classification methods. The formula created at this point can be defined as follows. In the formula below, $\hat{y}$ when the y state is 1 and $1-\hat{y}$ when 0 are essential in the measurement of uncertainty.

$$H(p,q) = -\sum_i p_i\log q_i = -y\log \hat{y} - (1-y)\log(1-\hat{y}) \tag{16}$$

B. Additional Entropy Derivatives Used in Artificial Intelligence

The accuracy and quality of the information should be examined in artificial intelligence systems when information and its quality are crucial. Entropy is one of the approaches available at the moment, even though there are many more [29]. The derivatives of entropy, which are seen in many ways in artificial intelligence systems, are discussed in this section.

1) Maximum Entropy

Finding the best distribution to model on a dataset we already know can be crucial. Considering the earlier information, it can be concluded that the distribution that offers the most entropy among them is the best suitable in this situation. It is mainly utilized in Bayesian inference-based methodologies. While the fundamental formula is shown below, the essential objective is identifying the distribution that optimizes entropy [30].

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$$S = -\sum_i p_t \log \frac{p_i}{m_i} \tag{17}$$

A theory known as Bayesian inference [31] is used to compare two random occurrences based on conditional probability. From this conclusion, several techniques have been developed, including Naive Bayes. Entropy is the foundation for inference, which operates based on information disorder on other information. The appropriate equation is shown below. P(A|B) is the conditional probability of A for B, and P(A) is the probability of A occurring alone.

$$P(A \mid B) = \frac{P(B|A)P(A)}{P(B)} \tag{18}$$

2) Information Gain

Decision trees are built via decision routing that begins at the root node and works downward. The tree is constructed using entropy. Entropy is the technique employed. Information Gain (IG) is the term for this entropy-based technique. On the other hand, information gain is made up of the entropy difference between states [32]. Decision tree techniques like ID3 and C4.5 utilize it. The following is a presentation of the Information Gain formula.

$$IG(S, A) = H(S) \& -\sum_{t \in T} p(t)H(t) = H(S) - H(S \mid A) \tag{19}$$

3) Loss Function

Loss functions are employed in classification techniques like Artificial Neural Networks to determine if a condition exists or not. The Mean Squared Error is the most common form among others. Contrary to the approach we discussed in cross-entropy, it is the average of the cross-entropy values gathered for n samples in this structure [33]. Evaluation and operations may then be made based on the average of the information's irregularity. The equation is shown below. The number of samples is represented by n, whereas the number of instances is represented by i.

$$L = -\frac{1}{n}\sum_{j=1}^{n} \sum_i p_i \log q_i \tag{20}$$

4) Logistic Regression

One of the fundamental classification methods is this one. It is the logarithm of the ratio between an event's occurrence and nonoccurrence [34]. The method's formula, which differs from linear regression in that it uses categorical dependent variables, is shown below.

$$\log\ it(p_i) = \log\left(\frac{p_i}{1-p_i}\right) \tag{21}$$

IV. Methods of Artificial Intelligence

People's basic behaviors are shaped by their intelligence and prior learning impulses. Studies are now being conducted to shape the job successfully [35]. They will utilize this information on their computers, which are better at storing information than people. The capacity of the human brain to shape events has not yet been wholly realized based on technology, despite a certain threshold being passed at the point reached today [36].

The dataset used to solve an issue for artificial intelligence should contain information from past attempts or problems. Depending on the structure and content of this dataset, several preprocessing techniques are applied. Fig. 2 illustrates two fundamental categories in which supervised and unsupervised learning are investigated about Artificial Intelligence approaches utilized to resolve many different challenges, particularly Machine Learning [37]. Three subcategories are created from these categories. Classification, clustering, and regression are these [38].

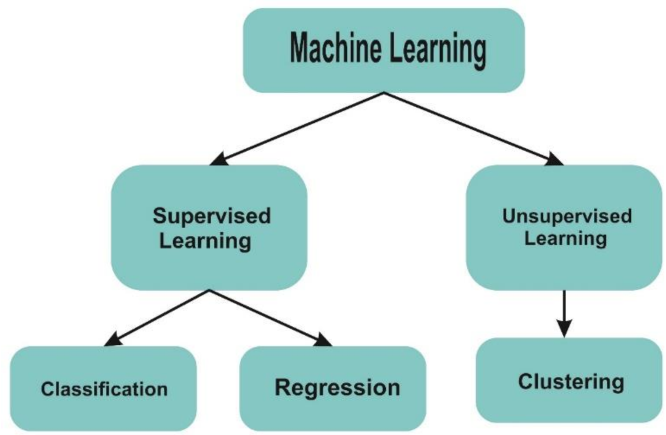

Fig. 2: Machine learning categorical structure supervise

First and foremost, the system has to be taught for categorization. The data provided during the training phase should express a specific outcome. The approach generates its model depending on the other parameters supplied and the outcome it expresses. Following that, each difficulty question is addressed utilizing knowledge gained throughout training [39]. When clustering, it is assumed that the provided information will be appropriately categorized. To sum up, the distinction in the distribution of the provided information is crucial in regression approaches. The main challenge is how to effectively define this distinction with the fewest possible false outcomes [40].

V. Shannon Entropy Applications

Entropy measures uncertainty while calculating the value of a random variable, according to Shannon's axioms. For instance, the outcome of a game of coin tossing is less predictable than the outcome of a game of dice [41]. The dice game has six equally likely outcomes, whereas the coin toss has two equally likely possibilities. As a result, there is less ambiguity in the coin flip game than in the dice game. Because of this, the coin toss game's uncertainty is less entropic than the dice games. Say we have a fair coin. The data in the Table 1 below is what we get when we flip our coin in a coin toss game [42].

TABLE 1: Fair Money Coin Flip Game

Outputs	Writing	Head-to-Head
Possibility	1/2	1/2
Number of Bits	1	

Info: -log2p	1	1
Entropy: p.log2p	1/2	1/2
Total Entropy	1	

Given that the likelihood of both heads and tails in this table equals 1/2 and 1, just one bit is required to indicate both outcomes. These occurrences all have an equal chance of happening. Hence the total amount of information they contain is 2. Entropy as a whole is 1.

Let us now consider if the coin we have is fake (that is, the case where the probabilities are not equal to each other).

TABLE 2: Coin Flip Game with Unfair Money

Outputs	Writing	Head-to-Head
Possibility	1/4	3/4
Number of Bits	1	
Info: -log2p	2	0.415
Entropy: p.log2p	0.5	0.307
Total Entropy	0.807	

According to Shannon's definition, entropy is a numerical value that may be expressed in specific units (bit, Napier, decibel) [43]. As a result, the entropy of a random process may be determined numerically and stated in the units above when the probability structure of the process is known. We must use the fewest possible bits while storing data on our systems. In this sense, the idea of entropy is beneficial to us. In the example shown below, the amount of information gain (bits) that may be acquired by employing entropy is analyzed by comparing the two techniques using Hartley's information formula. It was determined whether the option could save a question faster and with fewer bits after specific queries were asked of both approaches [44].

Assume that eight competitors advance to the final round of a running competition in which 50 participants with numbers ranging from 1 to 50 compete. These eight competitors will then be positioned on the running field in numerical order for the final round. Assume that each contender will compete on eight parallel running tracks in a simultaneous race. Our goal is to determine which of the final's competitors, number 25, will participate. We know that the participants will be lined up on the track in numerical order and that competitor number 25 will participate in the final round. The answer to the random questions will reveal the competitor's track, which is the required knowledge in this situation [45]. These questions will only receive a "yes" or "no" response. The table 3 below displays the finalist's numbers and songs.

Table 3: Numbers and Tracks of the Competitors

	Track 1	Track 2	Track 3	Track 4	Track 5	Track 6	Track 7	Track 8
Competitor Number	3	6	8	17	25	32	43	50

Let us think about 2 alternative approaches here.

Technique:

- Does the first track include competitor number 25.
- Competitor number 25 is in the second track.
- The third track is where competitor number 25 is located.
- The fourth track is where competitor number 25 is located.
- Inquiry ...

Strategy

- **Q 1:** Competitor number 25 is on track 4, correct? If so, the objective has been attained. If the answer is no, the tracks on the left and 17 will be removed because 25 is more than 17?
- **Q 2:** Competitor 25 is on the sixth track. If so, the objective has been attained. If the answer is no, the tracks on the right are dropped because 25 is fewer than 32?
- **Q 3:** The 25th competitor is on track 5. Indeed, "yes" will be the response?

If approach 1 is employed, it is possible to determine the competitor's track in the first question, but there is a 1/8 chance that this will occur, and there is a 7/8 chance that the proper track will not be selected. Assume that the initial query does not yield the racer's track. The possibility of finding it in the second question is 1/7, and with a probability of 6/7, the money may not be on this route. The solution, which is the knowledge of which track the competitor is on, is achieved after the seventh question if the rival's track cannot be identified after the first six questions. On the other hand, if the second strategy is chosen, the worst-case scenario is that the necessary information cannot be obtained in the first question, the desired information cannot be obtained after the second question, but the necessary information is unquestionably obtained after the third question [46].

One of the often-chosen techniques for grouping and estimation issues is decision trees. In the training phase, these algorithms provide a top-down or general-to-specific approach. This technique creates branches based on measuring the attribute value of each node in the tree structure, which resembles a flowchart [47]. Fig. 3 depicts a standard decision tree construction.

The root refers to the earliest cells in decision trees (root or root node). According to the root condition, each observation is assigned a "Yes" or "No" classification. Root cell nodes are present (interval nodes or nodes) [47]. Nodes are used to categorize observations. The complexity of the model increases with the number of nodes. The leaves (also known as leaf nodes or leaves) at the bottom of the decision tree provide the outcome. The root attribute is chosen using the entropy approach during tree building [48]. Each attribute's entropy value is determined using Shannon's formula, and information gain is derived by deducting these values from the system's overall entropy. Consider T as an attribute and D as a label, for instance. Entropy is defined as: In the case of binary progression (yes/no) [49].

$$\text{Entropy (D)} = -p_{yes}\left(\log_2 p_{yes}\right) - p_{no}\left(\log_2 p_{no}\right) \quad (22)$$

Here, $Pyes$ stand for the percentage of affirmative responses to the property, and Pno for the percentage of negative responses, it does. The weighted sum of the entropies in the subsets may then be used to compute the information needed to divide the T attribute into n segments in the D tag. The following is how information computations are expressed:

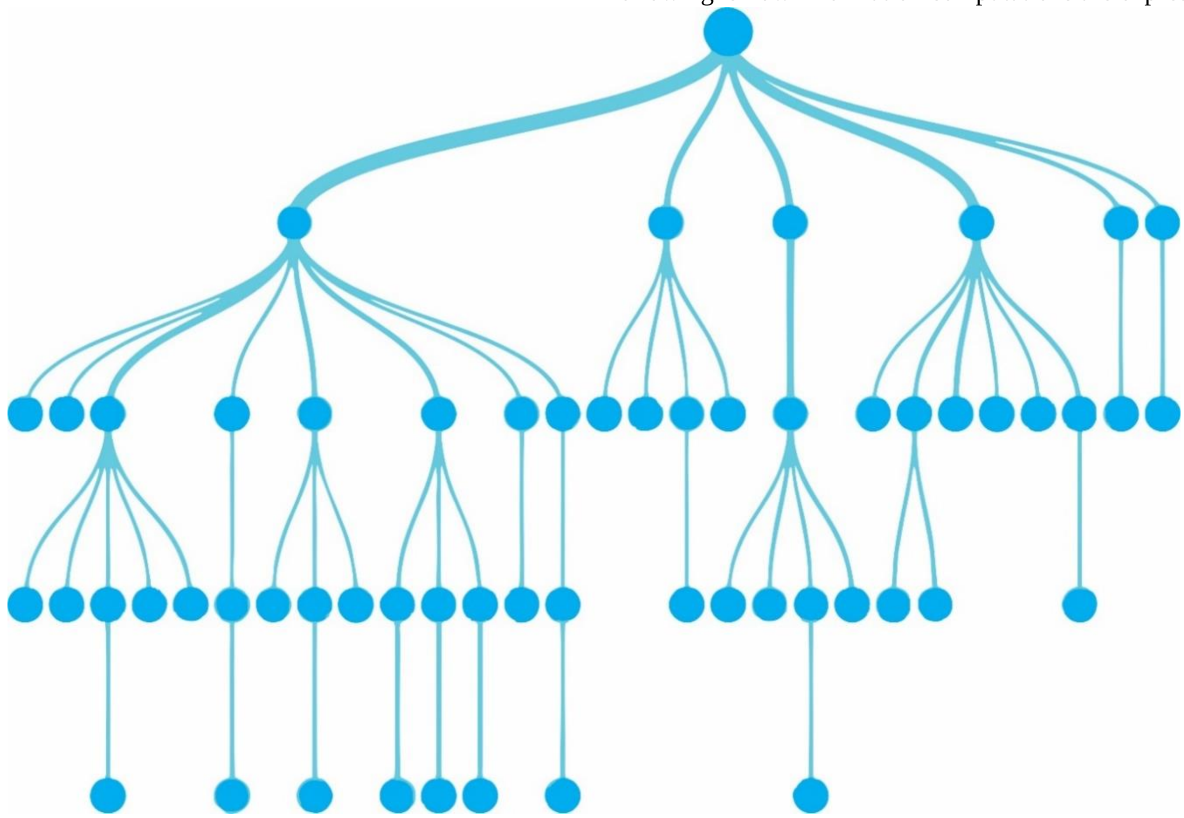

Fig. 3: Example Decision Tree

$$\text{Info}_T\text{ (D)} = \sum_{i=1}^{n} \frac{Di}{D} \text{Entropy } (D_i) \quad (23)$$

Last but not least, the information gain from branching on the T property may be computed using:

$$\text{Gain (T)} = \text{ Entropy (D) } - \text{Info}_T\text{ (D)} \quad (24)$$

The model is then generated in a top-down recursive divide-and-rule tree using the characteristics with the largest information gain. The existing data set is split into training and test sets in order to build a model. Let's use an example data set and the approach described above to design a decision tree [40]. Let's make a decision tree and assess the employment status of potential employees in light of the three attributes listed. Table 4 contains the sample dataset:

TABLE 4: Recruitment by Qualifications

Age (<30, 30-40, >30)	**Foreign Language Proficiency**	**Salary expectation (Less than 5000 $, more)**	**Hiring Status (Yes, No)**
<30	Sufficient	Little	No
<30	Sufficient	More	No
30-40	Sufficient	Little	Yes
>40	Sufficient	Little	Yes
>40	Insufficient	Little	Yes
>40	Insufficient	More	No
30-40	Insufficient	More	Yes
<30	Sufficient	Little	No
<30	Insufficient	Little	Yes
>40	Insufficient	Little	Yes
<30	Insufficient	More	Yes
30-40	Sufficient	More	Yes
30-40	Insufficient	Little	Yes

>40	Sufficient	More	No

The entropy method determines which factors (age, foreign language competency, and pay expectation) should be given greater weight when evaluating candidates. Entropy is determined for each characteristic individually, and information gain is discovered. The tree's root determines which gain is the greatest. If a sample computation is performed for the "Age" property, for instance:

TABLE 5: Age Criteria Information Amount

Age	Yes	No	Info (Yes, No)
<30	2	3	I (2,3)
30-40	4	0	I (4,0)
>40	3	2	I (3,2)

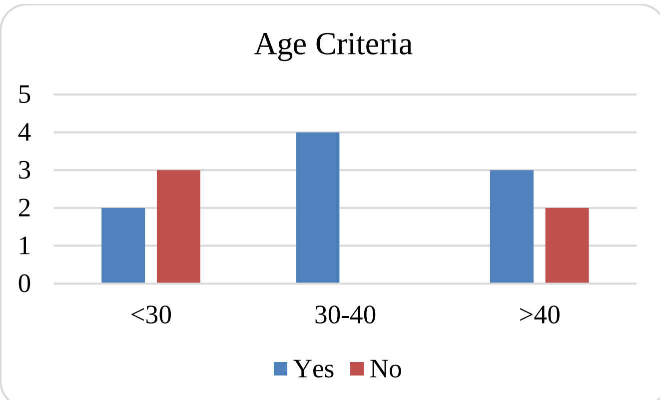

Fig. 4: Age Criteria Information Amount

The system's entropy value is determined using Eq. 22, as shown below.

$$\text{Entropy } (age) = I(9{,}5)$$

$$= -\frac{9}{14}\log_2\frac{9}{14} - \frac{5}{14}\log_2\frac{5}{14}$$

$$= 0.940,$$

The amount of data in the age criteria is determined using Eq. 23, as shown below.

$$\text{Info}_{\text{age}}\text{ (work} = \frac{5}{14}\text{I}(2{,}3) + \frac{4}{14}\text{I}(4{,}0) + \frac{5}{14}\text{I}(3{,}2)$$

$$=.\frac{5}{14}\left(-\frac{2}{5}\log_2\frac{2}{5} - \frac{3}{5}\log_2\frac{3}{5}\right) + \frac{4}{14}\left(-\frac{4}{4}\log_2\frac{4}{4} - \frac{0}{4}\log_2\frac{0}{4}\right) + \frac{5}{14}\left(-\frac{3}{5}\log_2\frac{3}{5}\frac{2}{5}\log_2\frac{2}{5}\right)$$

$$= 0.694,$$

The following Eq. 24, is used to determine the quantity of information obtained.

$$\begin{aligned}\text{Gain (age)} &= \text{Entropi (work)} - \text{Info}_{\text{age}}\text{ (work)} \\ &= 0{,}940 - 0{,}694 \\ &= 0.246.\end{aligned}$$

The increases for other qualities' entropy and information are calculated similarly: Gain (Foreign language proficiency) = 0.151, Gain (Salary expectation) = 0.048. Starting the required decision tree according to the quality output with the most significant information gain, namely the age criterion, would be desirable. Fig. 5 depicts the decision tree produced by the findings attained.

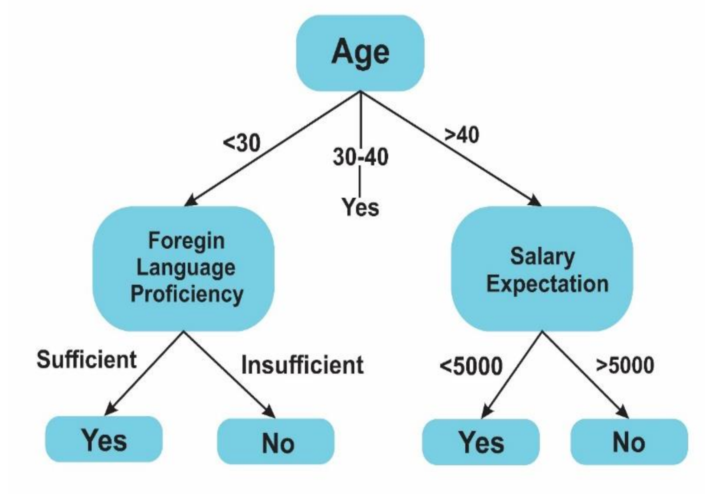

Fig. 5: Result Decision Tree

VI. CONCLUSION

Today, knowledge is valued most. Raw data needs to be more. It must be understood and valued. Measurable is the first requirement. This paper discusses entropy, a fundamental approach to quantifying information's value from all aspects. The presentation and debate on Artificial Intelligence (AI) may use, one of the study's primary issues, will help improve AI methodologies and application models. Entropy is a probabilistic measure of information, but AI uses it based on problem spaces. Since categorizing an item with specified qualities is an issue, cross-entropy is typically found. When considering solutions to the provided challenges, the goal is to get the system to the most logical and correct outcome by minimizing information loss from the mathematical model produced during method development. Measure the disorder of entropy-based information in this structure. Entropy, the measure of information in information theory, indicates that less probable occurrences contain more information and more probable events contain less. When the odds of two occurrences are comparable, the result is more unknown. Hence entropy is the highest. Hartley's definition and Shannon's formulations of entropy expose challenges in information technology, notably in information transmission, assuring the proper transfer of information in the quickest method, and making judgments in data storage and artificial learning with fewer bits. It helps produce trees. This notion has helped advance IT. In this paper, entropy-based AI algorithms were given from a new angle. Entropy is applied in every sector that requires information and interpretation. A future study will examine the usefulness of different entropy in AI algorithms and their performance.

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